

Series TD1: (basic concepts)

Questions :

1. What are the principal components of computer?
2. Explain the difference between volatile and non-volatile memory.
3. What are the main roles of the central processing unit (CPU) in a computer?
4. What is a data bus and how does it work?
5. Describe the main functions of the following components: processor, RAM, ROM and hard disk.
6. What is the difference between an input device and an output device? Give examples.
7. What is a motherboard and what are its main functions?
8. What are the characteristics of Von Neumann architecture?
9. What is an instruction cycle and what are its steps?
10. What is a binary system and why is it used in computers?
11. How many states can be represented with 3 bits? with 4 bits? and with n bits in general?
12. How to write a number a in base a ?
13. Using three bits, show the binary counting sequence from 000 to 111.
14. What is the maximum number that we can count up to using 10 bits?
15. How many bits are needed to count up to a maximum of 63?
16. What is the largest decimal value that can be represented by:
 - (a) an eight-bit binary number?
 - (b) A 16-bit number?

Exercise1: Specify the units of measurement in the following data sheet:

- Intel Core™i5 (frequency 3.40, cache memory 4)
- Windows 8.1 64
- RAM4.....with frequency of 1333
- Hard disk 850, transfer rate 4
- Integrated network card (LAN) : 100
- ADSL connection 2
- WebCam : resolution 12

Exercise2: Perform the conversions to the appropriate units for the following values:

$$300 \text{ MHz} = ? \text{ GHz}$$

$$2,89 \text{ GHz} = ? \text{ KHz}$$

$$1,8 \text{ GHz} = ? \text{ Hz}$$

$$500 \text{ Go} = ? \text{ To}$$

$$3072 \text{ Mo} = ? \text{ Go}$$

$$50 \text{ o} = ? \text{ Bit}$$

$$4 \text{ Ko} = ? \text{ Bit}$$

$$2^{30} \text{ Bit} = ? \text{ Mo}$$

$$1 \text{ To} = ? \text{ Ko}$$

Exercise3: Express each of the given numbers in polynomial form based on their respective bases:

$$(101101)_2$$

$$(1257)_{10}$$

$$(76)_8$$

$$(F15B)_{16}$$

Additional exercise to look up:

Exercise1:

A student wants to transfer software to a storage unit such as:

The software occupies 222 KB.

The USB flash drive has a capacity of 8 GB.

The CD's capacity is 1400 MB.

- 1- Is it possible to store the software on the USB flash drive? Explain.
- 2- How many CDs are needed to store the software?
- 3- What can you deduce?

Exercise2:

- 1- Find the bases x and y such that: $(403)_x = (103)_{10}$, $(147)_y = (103)_{10}$
- 2- Find the base b
 1. $(1001)_b = (201)_{16}$
 2. $(1001100)_b = (114)_8$
 3. $(125)_b = (293)_{10}$